

Solution Sheet-05

1.

(a) Factorization Theorem for Independence

Statement: Let (X, Y) have joint density f . Show that X and Y are independent if and only if there exists a constant $k > 0$ and nonnegative functions f_1 and f_2 such that:

$$f(x, y) = k f_1(x) f_2(y)$$

for all $x, y \in \mathbb{R}$.

Proof:

Direction 1 ($\Rightarrow$): Assume X and Y are independent. By the definition of independence for continuous random variables, the joint PDF is the product of the marginal PDFs:

$$f(x, y) = f_X(x) f_Y(y)$$

This equation satisfies the required form $f(x, y) = k f_1(x) f_2(y)$ by choosing:

- $k = 1$
- $f_1(x) = f_X(x)$
- $f_2(y) = f_Y(y)$

Thus, the condition holds.

Direction 2 ($\Leftarrow$): Assume $f(x, y) = k f_1(x) f_2(y)$. We must show that $f(x, y) = f_X(x) f_Y(y)$. First, let's define the integrals of the functions f_1 and f_2 :

$$C_1 = \int_{-\infty}^{\infty} f_1(x) dx \quad \text{and} \quad C_2 = \int_{-\infty}^{\infty} f_2(y) dy$$

Since $f(x, y)$ is a valid PDF, it integrates to 1 over the whole plane:

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} k f_1(x) f_2(y) dx dy &= 1 \\ k \left(\int_{-\infty}^{\infty} f_1(x) dx \right) \left(\int_{-\infty}^{\infty} f_2(y) dy \right) &= 1 \\ k C_1 C_2 = 1 &\implies k = \frac{1}{C_1 C_2} \end{aligned}$$

Next, we find the marginal densities $f_X(x)$ and $f_Y(y)$ by integrating the joint density:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^{\infty} k f_1(x) f_2(y) dy = k f_1(x) \int_{-\infty}^{\infty} f_2(y) dy \\ f_X(x) &= k f_1(x) C_2 \end{aligned}$$

Similarly for Y :

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{-\infty}^{\infty} k f_1(x) f_2(y) dx = k f_2(y) \int_{-\infty}^{\infty} f_1(x) dx$$

$$f_Y(y) = k f_2(y) C_1$$

Now, check the product of the marginals:

$$\begin{aligned} f_X(x) f_Y(y) &= (k C_2 f_1(x)) \cdot (k C_1 f_2(y)) \\ &= k(k C_1 C_2) f_1(x) f_2(y) \end{aligned}$$

Using the fact that $k C_1 C_2 = 1$:

$$= k(1) f_1(x) f_2(y) = f(x, y)$$

Since $f(x, y) = f_X(x) f_Y(y)$, X and Y are independent. □

(b) Rectangular Support Property

Statement: Let $A = \{x : f_X(x) > 0\}$ and $B = \{y : f_Y(y) > 0\}$. Show that if X and Y are independent, then:

$$\{(x, y) : f(x, y) > 0\} = A \times B$$

Proof: Let S be the support of the joint distribution, defined as $S = \{(x, y) : f(x, y) > 0\}$. We are given that X and Y are independent, which implies:

$$f(x, y) = f_X(x) f_Y(y) \quad \text{for all } (x, y).$$

We need to prove that $(x, y) \in S \iff (x, y) \in A \times B$.

Step 1: From Joint Support to Product Support Let $(x, y) \in S$. By definition, this means $f(x, y) > 0$. Substituting the independence condition:

$$f_X(x) f_Y(y) > 0$$

Since probability densities are non-negative ($f_X(x) \geq 0$ and $f_Y(y) \geq 0$), their product is strictly positive if and only if **both** terms are strictly positive.

$$f_X(x) > 0 \quad \text{AND} \quad f_Y(y) > 0$$

By the definitions of sets A and B :

$$x \in A \quad \text{AND} \quad y \in B$$

This is the definition of the Cartesian product:

$$(x, y) \in A \times B$$

Step 2: From Product Support to Joint Support Let $(x, y) \in A \times B$. This means $x \in A$ (so $f_X(x) > 0$) and $y \in B$ (so $f_Y(y) > 0$). Multiplying these two positive numbers:

$$f_X(x) f_Y(y) > 0$$

Using the independence condition $f(x, y) = f_X(x) f_Y(y)$:

$$f(x, y) > 0$$

Therefore, $(x, y) \in S$.

Conclusion: Since the implications work in both directions, $S = A \times B$. (This explains why non-rectangular

2. Conditional Expectation on Dependent Support

Problem: $f(x, y) = 2$ on $0 < x < y < 1$.

Solution: Conditional PDF $f_{X|Y}(x|y)$: First, find the marginal PDF of Y . For a fixed $y \in (0, 1)$, x ranges from 0 to y :

$$f_Y(y) = \int_0^y f(x, y) dx = \int_0^y 2 dx = 2y, \quad 0 < y < 1.$$

Now use the definition

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2}{2y} = \frac{1}{y}, \quad 0 < x < y.$$

This indicates that given $Y = y$, X is **Uniformly distributed** on $(0, y)$.

3. The Factorization Trap (Diamond Region)

Problem: $f(x, y) = 1/2$ on $|x| + |y| \leq 1$.

Solution:

- (a) **Marginal PDFs:** The region is a square rotated by 45° . For a fixed x (where $-1 \leq x \leq 1$), the condition $|x| + |y| \leq 1$ implies $|y| \leq 1 - |x|$, so $-(1 - |x|) \leq y \leq (1 - |x|)$.

$$f_X(x) = \int_{-(1-|x|)}^{1-|x|} \frac{1}{2} dy = \frac{1}{2} \times \text{Length} = \frac{1}{2} \cdot 2(1 - |x|) = 1 - |x|.$$

Similarly, by symmetry, $f_Y(y) = 1 - |y|$ for $-1 \leq y \leq 1$.

- (b) **Independence Check:** If independent, $f(x, y) = f_X(x)f_Y(y)$ must hold for all x, y . Consider the point $(0, 0)$:

$$\text{LHS} = f(0, 0) = 1/2.$$

$$\text{RHS} = f_X(0)f_Y(0) = (1 - |0|)(1 - |0|) = 1 \times 1 = 1.$$

Since $1/2 \neq 1$, factorization fails. **Not Independent.**

- (c) **Explanation:** Even though the density function $f(x, y) = 1/2$ is constant (which looks like it factors as $C \times 1 \times 1$), the **domain of support** cannot be written as a Cartesian product $A \times B$. The range of Y depends on X . Dependence is "coupled" through the boundary constraints.

4. Pairwise vs. Mutual Independence

Problem: X_1, X_2 coin tosses (0 or 1 with prob 0.5). $X_3 = |X_1 - X_2|$.

Solution: First, find the distribution of X_3 : $X_3 = 0$ if $(X_1, X_2) \in \{(0, 0), (1, 1)\}$ (Prob $0.25 + 0.25 = 0.5$). $X_3 = 1$ if $(X_1, X_2) \in \{(0, 1), (1, 0)\}$ (Prob 0.5). So X_1, X_2, X_3 are all Bernoulli(0.5).

- (a) **Pairwise (X_1, X_3) :** Check $P(X_1 = 0, X_3 = 0)$. This requires $X_1 = 0$ and $|0 - X_2| = 0 \implies X_2 = 0$. $P(X_1 = 0, X_2 = 0) = 1/4$. Compare to product: $P(X_1 = 0)P(X_3 = 0) = (1/2)(1/2) = 1/4$. This holds for all combinations. They are independent.
- (b) **Pairwise (X_2, X_3) :** Symmetric to part (a). Independent.

- (c) **Mutual Independence:** Check the joint probability $P(X_1 = 0, X_2 = 0, X_3 = 1)$. If $X_1 = 0$ and $X_2 = 0$, then $X_3 = |0 - 0| = 0$. Thus, the event $(0, 0, 1)$ is **impossible**.

$$P(X_1 = 0, X_2 = 0, X_3 = 1) = 0.$$

However, the product of marginals is:

$$P(X_1 = 0)P(X_2 = 0)P(X_3 = 1) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $0 \neq 1/8$, is violated. They are **not mutually independent**.

5. Transformations of Uniform Random Variables

Let X and Y be i.i.d. random variables with PDF $f(x) = 1$ for $0 < x < 1$ (Uniform distribution $U(0, 1)$). The joint PDF is $f_{X,Y}(x, y) = 1$ for $0 < x, y < 1$.

- (a) **PDF of $Z = XY$:** The range of Z is $(0, 1)$. The CDF is $F_Z(z) = P(XY \leq z) = \iint_{xy \leq z} 1 \, dx \, dy$. We integrate over the region defined by $y \leq z/x$ within the unit square. The intersection of the curve $y = z/x$ and $y = 1$ occurs at $x = z$.

$$F_Z(z) = \int_0^z \int_0^1 dy \, dx + \int_z^1 \int_0^{z/x} dy \, dx$$

$$F_Z(z) = \int_0^z 1 \, dx + \int_z^1 \frac{z}{x} \, dx = z + z[\ln x]_z^1 = z + z(0 - \ln z) = z - z \ln z.$$

Differentiating with respect to z :

$$f_Z(z) = \frac{d}{dz}(z - z \ln z) = 1 - (1 \cdot \ln z + z \cdot \frac{1}{z}) = 1 - \ln z - 1 = -\ln z.$$

Answer: $f_{XY}(z) = \begin{cases} -\ln z, & 0 < z < 1 \\ 0, & \text{otherwise} \end{cases}$.

- (b) **PDF of $Z = X/Y$:** The range of Z is $(0, \infty)$. CDF $F_Z(z) = P(X/Y \leq z) = P(X \leq zY)$.

- Case $0 < z \leq 1$: The region is a triangle with vertices $(0, 0)$, $(z, 1)$, $(0, 1)$.

$$F_Z(z) = \int_0^1 \int_0^{zy} dx \, dy = \int_0^1 zy \, dy = \frac{z}{2}.$$

- Case $z > 1$: The region is the complement of the triangle where $X > zY$ (vertices $(0, 0)$, $(1, 0)$, $(1, 1/z)$).

$$F_Z(z) = 1 - P(X > zY) = 1 - \int_0^1 \int_0^{x/z} dy \, dx = 1 - \int_0^1 \frac{x}{z} \, dx = 1 - \frac{1}{2z}.$$

Differentiating:

$$f_Z(z) = \begin{cases} \frac{1}{2}, & 0 < z \leq 1 \\ \frac{1}{2z^2}, & z > 1 \end{cases}.$$

- (c) **PDF of $U = \min\{X, Y\}$:** $P(U > u) = P(X > u, Y > u) = P(X > u)P(Y > u) = (1 - u)^2$ for $0 < u < 1$. CDF $F_U(u) = 1 - (1 - u)^2$. PDF $f_U(u) = 2(1 - u)$ for $0 < u < 1$.

- (d) **PDF of $V = \max\{X, Y\}$:** CDF $F_V(v) = P(V \leq v) = P(X \leq v, Y \leq v) = P(X \leq v)P(Y \leq v) = v^2$ for $0 < v < 1$. PDF $f_V(v) = 2v$ for $0 < v < 1$.

6. Sum of Three Uniform Variables

Let $X_i \sim U(0, 1)$. We find the PDF of $U = X_1 + X_2 + X_3$ by convolution.

Step 1: Sum of two variables $S_2 = X_1 + X_2$ The PDF $f_{S_2}(s)$ is the convolution of two boxcar functions. This yields the triangular distribution:

$$f_{S_2}(s) = \begin{cases} s, & 0 < s < 1 \\ 2 - s, & 1 \leq s < 2 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Sum of three variables $U = S_2 + X_3$

$$g(u) = \int_{-\infty}^{\infty} f_{S_2}(y) f_{X_3}(u - y) dy = \int_{u-1}^u f_{S_2}(y) dy$$

where the limits come from $0 < u - y < 1 \implies u - 1 < y < u$. We intersect this window $[u - 1, u]$ with the support of f_{S_2} ($[0, 2]$).

- **Region 1** ($0 < u < 1$): Integration range is $[0, u]$.

$$g(u) = \int_0^u y dy = \frac{u^2}{2}.$$

- **Region 2** ($1 < u < 2$): The window $[u - 1, u]$ overlaps with both parts of the triangle. Split at $y = 1$.

$$\begin{aligned} g(u) &= \int_{u-1}^1 y dy + \int_1^u (2 - y) dy \\ &= \left[\frac{y^2}{2} \right]_{u-1}^1 + \left[2y - \frac{y^2}{2} \right]_1^u \\ &= \left(\frac{1}{2} - \frac{(u-1)^2}{2} \right) + \left((2u - \frac{u^2}{2}) - (2 - \frac{1}{2}) \right) \\ &= \frac{1}{2} - \frac{u^2 - 2u + 1}{2} + 2u - \frac{u^2}{2} - \frac{3}{2} \\ &= -u^2 + 3u - \frac{3}{2}. \end{aligned}$$

- **Region 3** ($2 < u < 3$): Integration range is $[u - 1, 2]$.

$$\begin{aligned} g(u) &= \int_{u-1}^2 (2 - y) dy = - \left[\frac{(2 - y)^2}{2} \right]_{u-1}^2 \\ &= - \left(0 - \frac{(2 - (u - 1))^2}{2} \right) = \frac{(3 - u)^2}{2} = \frac{(u - 3)^2}{2}. \end{aligned}$$

Combining these gives the required formula.

7. Moment Generating Function and Independence

$$f(x, y) = \frac{1}{4}[1 + xy(x^2 - y^2)] \text{ for } |x| \leq 1, |y| \leq 1.$$

1. Independence Check: Find the marginal PDF of X :

$$f_X(x) = \int_{-1}^1 \frac{1}{4}[1 + x^3y - xy^3]dy = \frac{1}{4} \left[y + x^3 \frac{y^2}{2} - x \frac{y^4}{4} \right]_{-1}^1$$

The terms with even powers of y (y^2, y^4) vanish when evaluating limits -1 to 1 . The term y gives $1 - (-1) = 2$.

$$f_X(x) = \frac{1}{4}[2 + 0 - 0] = \frac{1}{2}, \quad -1 \leq x \leq 1.$$

Similarly, $f_Y(y) = \frac{1}{2}$. Both are Uniform($-1, 1$). Check product: $f_X(x)f_Y(y) = \frac{1}{4}$. Since $f(x, y) \neq \frac{1}{4}$ (it has the extra term $xy(x^2 - y^2)$), X and Y are **Dependent**.

2. Covariance: Since $X, Y \sim U(-1, 1)$, $E[X] = 0$ and $E[Y] = 0$. Thus $\text{Cov}(X, Y) = E[XY]$.

$$\begin{aligned} E[XY] &= \int_{-1}^1 \int_{-1}^1 xy \cdot \frac{1}{4}[1 + x^3y - xy^3]dxdy \\ &= \frac{1}{4} \left(\iint xy dA + \iint x^4y^2 dA - \iint x^2y^4 dA \right) \end{aligned}$$

$$\iint xy = (\int x)(\int y) = 0. \quad \iint x^4y^2 = (\int_{-1}^1 x^4 dx)(\int_{-1}^1 y^2 dy) = \left[\frac{x^5}{5}\right]_{-1}^1 \left[\frac{y^3}{3}\right]_{-1}^1 = \frac{2}{5} \cdot \frac{2}{3} = \frac{4}{15}. \quad \iint x^2y^4 = \frac{4}{15} \text{ (by symmetry).}$$

$$E[XY] = \frac{1}{4} \left(0 + \frac{4}{15} - \frac{4}{15} \right) = 0.$$

Answer: $\text{Cov}(X, Y) = 0$. (Uncorrelated but Dependent).

3. Moment Generating Function:

$$M(t_1, t_2) = E[e^{t_1X + t_2Y}] = \frac{1}{4} \iint (1 + x^3y - xy^3)e^{t_1x}e^{t_2y}dxdy$$

This splits into three integrals. Let $I_n(t) = \int_{-1}^1 x^n e^{tx} dx$.

$$M(t_1, t_2) = \frac{1}{4} [I_0(t_1)I_0(t_2) + I_3(t_1)I_1(t_2) - I_1(t_1)I_3(t_2)]$$

Standard integral results: $I_0(t) = \frac{2 \sinh t}{t}$. $I_1(t) = \frac{d}{dt} I_0(t) = \frac{2(t \cosh t - \sinh t)}{t^2}$. $I_3(t) = \frac{d^3}{dt^3} I_0(t)$ (or via integration by parts). The first term $\frac{1}{4} I_0(t_1)I_0(t_2)$ is the product of MGFs of independent Uniforms. The remaining terms represent the deviation due to dependence.

8. Helmert Transformation (Normal Variables)

$X_i \sim N(0, 1)$ i.i.d. The transformation $Y = AX$ is linear.

$$\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

Let this matrix be A .

- (a) **Jacobian:** We check if A is orthogonal. Row 1 norm: $\frac{1}{2} + \frac{1}{2} = 1$. Orthogonal to Row 2, $\frac{1}{\sqrt{12}} - \frac{1}{\sqrt{12}} + 0 = 0$. Row 2 norm: $\frac{1}{6} + \frac{1}{6} + \frac{4}{6} = 1$. Row 3 norm: $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$. Rows are orthonormal. Thus A is an orthogonal matrix ($AA^T = I$). Since $Y = AX$, $X = A^{-1}Y = A^TY$. The Jacobian determinant $J = \det(A^{-1}) = \det(A^T) = \det(A)$. For orthogonal matrices, $\det(A)^2 = 1 \implies |\det(A)| = 1$. **Answer:** $|J| = 1$.

- (b) **Sum of Squares:** Since A is orthogonal, it preserves the dot product (norm).

$$\sum Y_i^2 = Y^TY = (AX)^T(AX) = X^TA^TAX = X^TIX = X^TX = \sum X_i^2.$$

Verified.

- (c) **Joint PDF of Y :** $f_X(x_1, x_2, x_3) = \frac{1}{(2\pi)^{3/2}} e^{-\frac{1}{2}\sum x_i^2}$. Using the transformation formula:

$$h(y_1, y_2, y_3) = f_X(x(y))|J| = \frac{1}{(2\pi)^{3/2}} e^{-\frac{1}{2}\sum x_i^2} \cdot 1$$

Substitute $\sum x_i^2 = \sum y_i^2$:

$$h(y_1, y_2, y_3) = \frac{1}{(\sqrt{2\pi})^3} \exp\left(-\frac{y_1^2 + y_2^2 + y_3^2}{2}\right).$$

- (d) **Conclusion:** The joint PDF $h(y_1, y_2, y_3)$ factors into the product of three standard normal densities:

$$h(y) = \left(\frac{1}{\sqrt{2\pi}} e^{-y_1^2/2}\right) \left(\frac{1}{\sqrt{2\pi}} e^{-y_2^2/2}\right) \left(\frac{1}{\sqrt{2\pi}} e^{-y_3^2/2}\right).$$

Therefore, Y_1, Y_2, Y_3 are **independent** standard normal random variables.